Google Cloud H2S

AI Builder Cup

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# AML Insight — evidence-backed AML investigations

Team / project name: Genesis (repository project name)

Team leader: not recorded in the codebase

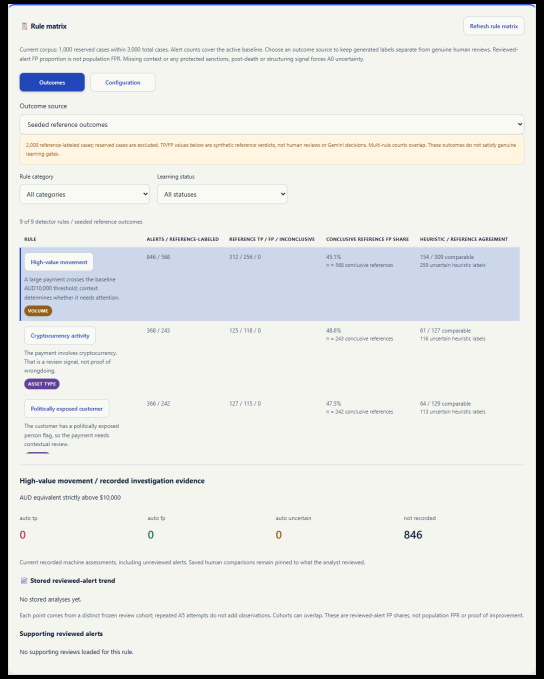
Problem statement: help analysts assess transaction alerts with traceable evidence, structured AI explanations and controlled learning from actual reviews.

# AML Insight / Genesis

A working investigation workspace for synthetic financial-crime cases.

Nine deterministic detectors
Six focused workflow roles
Evidence-linked Gemini assistance
Analyst-owned decisions

Current source: local prototype; earlier private cloud staging.




Actual local Evolution rule matrix
Seeded reference outcomes, not human verdicts.

# Brief about the idea

## Turn transaction alerts into evidence-linked investigations, then learn from genuine analyst outcomes.

### Investigate

Python enriches source facts and computes risk. Gemini explains, categorizes and writes a cited brief.

### Review

The analyst inspects source history, model findings and uncertainty, then saves a frozen verdict.

### Learn carefully

Evolution separates evidence sources and tests cutoff candidates before explicit application or rollback.

Current workspace: 3,000 synthetic cases / 200 identities / nine detector rules.

# Opportunities and proposed USP

## How it differs

Combines deterministic risk arithmetic with evidence-linked AI output, a sealed attempt ledger and a separate learning workflow.

No external competitor comparison is established by the code.

## How it solves the problem

Keeps source history, scores, counter-evidence and suggestions in one case view.

Structured validation rejects unsupported citation IDs and preserves uncertainty for analyst attention.

## Proposed USP

Traceable assistance with controlled adaptation: source labels, model opinions, drafts and human verdicts stay separate.

Python validates candidates; a model recommendation cannot apply itself.

Value proposition is implemented in the workflow; real-world productivity gains are not yet measured.

# Features offered by the solution

## Nine alert rules

High value, crypto, PEP, post-death, dormant, multi-currency, cash, foreign wire and structuring.

## Explainable risk

Five Python-owned dimensions: transaction 30%, customer 25%, account 15%, behavior 20%, network 10%.

## Case investigation

Identity/account links, source trail, risk explanation, typology, brief and agent traces.

## Durable execution

Parent/child runs, leases, checkpoints, deduplication and bounded retry handling.

## Evolution lab

Nine-rule matrix; separate reference, simulated and human outcomes; gated cutoff learning.

## Usage and access

Per-attempt tokens/cost status, daily controls, operator/admin allowlists and private outputs.

# Process flow / use case

Investigation: sequential stages, with source facts and risk owned by Python

**Source rules**
Transaction / alert

→

**A1**
Evidence enrichment

→

**A2**
Risk explanation

→

**A3**
Typology / action

→

**A4**
Cited case brief

→

**A0**
Priority / pre-label

Decision and learning: human feedback is required; generated labels do not qualify

**Analyst verdict**
Frozen feedback

→

**A5 Evolution**
Frozen evidence

→

**Python candidate**
Fit + validation

→

**Explicit approval**
Apply / observe / rollback

Missing context or protected signals require attention. AI output never becomes a human verdict.

# Wireframes / source-based screen map

## INVESTIGATIONS

- Dataset / coverage counters
- Identity + rule + status filters
- Paged investigation queue
- View details / reference
- Explicit triage preview

Read-only browsing; explicit actions.

## CASE DETAILS

- Identity / account / trigger
- Risk dimensions + evidence
- A2 → A3 → A4 → A0 trace
- Transactions / citations
- Saved analyst verdict

Private outputs require operator access.

## EVOLUTION LAB

- Outcomes / Understand / Learn
- Reference / simulated / human
- Nine-rule matrix + trends
- Candidate evidence + validation
- Apply / Observe / rollback

Simulation is labeled; no silent activation.

Schematic of existing components; actual prototype screenshots follow on slide 11.

# Architecture of the solution

**Browser**
React / TypeScript / Vite

→

**Cloud Run: web**
FastAPI / web + private API

→

**Firebase Auth**
Human sign-in / UID allowlists

Web ↔ Firestore • Web → Tasks → worker • Worker ↔ Firestore

**Firestore**
Sources / runs / reviews / versions

**Cloud Tasks**
Named jobs / OIDC delivery

→

**Cloud Run: worker**
Leases / stages / attempt ledger

Worker → Gemini via google-genai
Managed cloud identity; separate opt-in local Developer API path.

Cloud deployment design in code; earlier private staging exists, newest local source is not deployed.

# Technologies used in the solution

## Frontend

React 19, TypeScript, Vite; Firebase web SDK; accessible native dialogs.

## Backend

Python 3.12, FastAPI, Uvicorn and Pydantic; deterministic detection/scoring.

## Google AI

google-genai SDK; structured Gemini JSON, citations, per-leg model profiles and token budgets.

## Data / identity

Firestore, Firebase Authentication; emulator-backed local development.

## Deployment

Cloud Run web + worker; Cloud Tasks; Artifact Registry; Cloud Build; Secret Manager / ADC.

## Verification

pytest, Firebase emulator integration tests, TypeScript checks and Playwright browser journeys.

# Estimated implementation cost

## What the repository records

Earlier build: 118.60 seconds; recorded compute list-rate estimate US$0.01186, before free allowances. Not an invoice or a complete project cost.

## What is not established

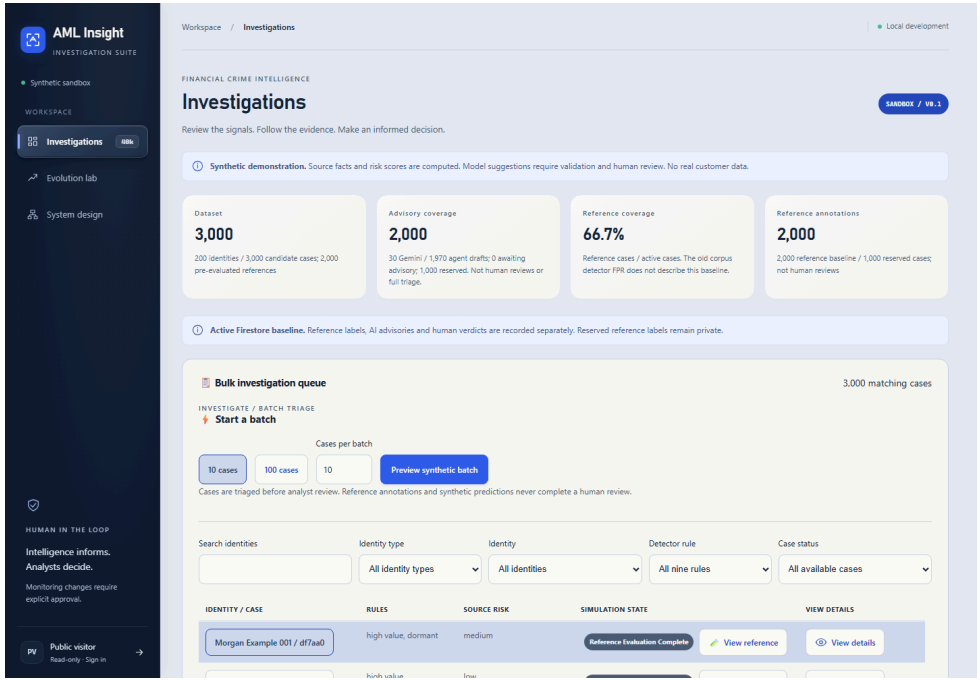
No labor-cost estimate or final monthly bill is recorded. Model cost remains UNKNOWN unless a matching verified rate card and usage are available.

## Configured token ceilings per investigation (before retries)

A2: 4,096 input / 1,536 generated • A3: 2,048 / 768 • A4: 4,096 / 2,048
Total A2–A4: 10,240 input / 4,352 generated. A5 is separate: 8,192 / 3,072.

Controls: scale 0–1 in recorded staging, daily attempt controls and paused dispatch.
The recorded AUD 6 alerts-only budget is not a spending cap; storage/network/build costs are separate.

# Snapshots of the prototype




Actual local queue: 3,000 cases / 200 identities.




Actual case-details dialog.
Synthetic customer data only.

# Prototype performance / benchmarking

**162**
unit tests passed

**25**
emulator integrations passed

**9**
detector rules covered

Recorded completed checkpoint: CI-012; 25 integrations, zero integration skips.

## What was exercised

Source integrity, scoped access, deduplication, projections and browser layouts from 320–1440 px. TypeScript and design checks passed.

## What is not a benchmark

No validated production accuracy, p95 latency, false-positive reduction or analyst-time saving is established. CI-013 is in progress.

These are recorded verification results, not tests rerun for this deck or evidence of deployed-current-source performance.

# Additional details / future development

## Complete the current workflow

Finish and verify the reserved-case genuine-batch work. Gather real analyst verdicts and measure outcomes; generated projections cannot satisfy learning gates.

## Validate improvement

Require 30 conclusive genuine reviews overall and 15 eligible target-rule observations; then prospective validation on five new distinct identities.

## Deploy current source

Build the verified current revision; validate publisher, Auth and private web/worker integration. Keep model/task activation explicit and bounded.

## Finish release artifacts

Publish clean source, a signed-out prototype and a real video under three minutes. Keep private notes, held-out labels and credentials out of release assets.

These are existing repository tasks, not new product promises or claimed completed capabilities.

# Repository, video and product links

## GitHub public repository

No published public repository URL is recorded. Clean source-export copies exist in submission/github-repository/.

## Demo video — under three minutes

No recorded video URL or measured duration is available. A 150-second recording script exists in submission/demo-video/.

## Final product link

Local source preview: http://127.0.0.1:5173 — this workstation only. Earlier private Cloud Run staging is recorded; no signed-out public product URL is recorded.

All three requested fields are answered from the repository; unavailable public links are not fabricated.

# Evidence boundaries: know what each count means

## 2,000 advisory cases: coverage is not the same as human verification



**30 genuine Gemini advisories (1.5%) + 1,970 agent drafts (98.5%).**

Agent drafts have distinct provenance and no provider attempts or human-feedback claims.
1,000 reserved cases are excluded from advisory preprocessing.

### Simulated analyst assignments

1,500 agree / 400 disagree / 100 uncertain. Generated training labels, not actual human verdicts.

### Current learning outcome

No supported cutoff candidate in the recorded projection run. No operational policy changed.

Reference labels, full investigations, Gemini advisories, agent drafts and human reviews remain separate.

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# Thank you



## AML Insight / Genesis

Evidence first.
AI assists. Analysts decide.
Learning requires validated outcomes.